

6	(a)	speed = Z/ρ		
		= $1.4 \times 10^6 / 940$ (=1490)	C1	
		time = $(1.1 \times 10^{-2} \times 2) / 1490$	C1	
		= $1.5 \times 10^{-5} \text{ s}$	A1	[3]

(time of $7.4 \times 10^{-6} \text{ s}$ is one way only and scores 2/3 marks)

(use of speed of light is wrong physics and scores 0/3 marks)

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- (b) $I = I_0 \exp(-\mu x)$ or $I_2 = I_1 \exp(-\mu x)$ C1
ratio = $\exp(-48 \times 1.1 \times 10^{-2})$
= 0.59 A1 [2]
- (c) $0.33/100 = 0.59 \times (I_3/I_2) \times 0.59$ C1
ratio = 9.5×10^{-3} A1
or
 $0.33/100 = \exp(-48 \times 2.2 \times 10^{-2}) \times (I_3/I_2)$ (C1)
ratio = 9.5×10^{-3} (A1) [2]
- (d) ratio I_3/I_2 increases B1 [1]
(accept: “there is an increase in the proportion of the intensity that is reflected”)